

Material Selection for a Microwave Turntable Plate

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1. Introduction

The microwave turntable plate is a critical component designed to enable uniform heating by rotating food placed in a microwave oven. This plate must maintain dimensional stability across repeated thermal cycles, resist failure from impact, and meet stringent food safety requirements. The material must also be microwave-transparent, meaning it must not reflect or absorb microwave radiation. The conventional material of choice is borosilicate glass, favored for its low thermal expansion and chemical inertness. However, its inherent brittleness and susceptibility to cracking make it less than ideal for long-term durability, especially under impact or thermal shock conditions. This project explores whether an alternative material can offer better performance or cost-effectiveness using Ashby's material selection methodology. Quantitative analysis is performed through the derivation of performance indices tailored to this application and visual screening using Ashby material selection charts.

2. Functional Requirements and Constraints

To identify appropriate candidate materials, we begin by defining the component's function and operating environment. The microwave turntable must support the weight of food and containers during rotation, remain stable under exposure to temperatures from room temperature to approximately 100°C, and withstand occasional impact such as accidental dropping during cleaning or handling. Additionally, the material must be non-toxic, chemically inert, and microwave-transparent, ruling out all conductive materials. Although friction between the plate and food containers was not included as a primary constraint, it is worth noting that sliding resistance is a two-body problem affected by both the turntable and the container. Surface texture or surface treatment could be used to locally increase friction if necessary, regardless of the base material.

From these requirements, we extract the following key constraints that candidate materials must satisfy:

- **Support a static load up to 5 kg** without permanent deformation

- **Withstand repeated thermal cycling** between 25°C and 100°C
- **Resist cracking** due to impact from drops
- **Remain chemically stable** and **food-safe**
- **Be non-metallic and microwave-transparent**
- **Be cost-effective** for mass production

These translate to the following quantitative property constraints:

- **Yield Strength:** $\sigma_y \geq 30 \text{ MPa}$
- **Fracture Toughness:** $K_{IC} \geq 0.8 \text{ MPa}\sqrt{m}$
- **Thermal Expansion:** $\alpha \leq 10 \times 10^{-6}/\text{K}$
- **Cost:** $C \leq 3 \text{ \$/kg}$
- **Microwave Transparency:** Must be dielectric (non-conductive)

3. Material Index Derivations

To perform quantitative comparisons among material candidates, we derive five performance indices corresponding to the five most critical objectives. These indices are developed from simplified mechanical and thermal models and capture the essential trade-offs between properties.

The first index is thermal stability, which is important for dimensional control under heat. When temperature changes cause thermal strain $\varepsilon = \alpha\Delta T$, this strain can result in stress $\sigma = E\alpha\Delta T$ when the material is constrained. Thus, to minimize dimensional changes or induced stress, the thermal stability index becomes:

$$M_1 = \frac{E}{\alpha}$$

The second index focuses on minimizing the weight of the plate while maintaining strength. This is especially relevant for ease of handling, cost of materials, and rotational inertia. From basic mechanics, the strength-to-weight performance index is:

$$M_2 = \frac{\sigma_y}{\rho}$$

The third index measures impact resistance. From fracture mechanics, the energy required to propagate a crack is proportional to K_{IC}^2/E , where K_{IC} is the fracture toughness. This leads to the performance index:

$$M_3 = \frac{K_{IC}^2}{E}$$

The fourth index relates to thermal shock resistance. A sudden temperature change induces thermal stress on the order of $E\alpha\Delta T$. Cracking occurs when this stress exceeds the material’s toughness, leading to the composite index:

$$M_4 = \frac{K_{IC}}{E\alpha}$$

Finally, cost-performance is a major factor in mass-produced components. Assuming strength is the limiting factor under normal use, the cost efficiency index is:

$$M_5 = \frac{\sigma_y}{C}$$

Each index will be used to screen and rank candidate materials in the following sections.

4. Candidate Materials and Properties

Based on the functional requirements and material indices derived earlier, seven candidate materials were selected for evaluation. The selection includes traditional ceramics, technical polymers, and high-performance glasses.

Borosilicate glass is the current material used in microwave turntables and serves as our baseline. It is chemically inert, thermally stable, and microwave transparent, but has limited fracture toughness. Fused silica was chosen for its extremely low thermal expansion, making it an ideal benchmark for thermal stability, though it is known to be brittle. Soda-lime glass, a lower-cost alternative to borosilicate, is evaluated for comparison but is expected to underperform in thermal stability.

Polycarbonate (PC) is included for its high fracture toughness and ductility, offering potential improvements in impact resistance. Polysulfone (PSU), a higher-performance polymer than PC, was selected for its balance of strength, toughness, and temperature resistance. PEEK (polyether ether ketone) is a high-end engineering thermoplastic with excellent toughness and thermal performance but is typically cost-limiting for mass production.

Alumina (Al_2O_3) was added as a high-performance technical ceramic with outstanding stiffness and thermal resistance. Although it is not microwave-transparent and exceeds the cost limit, it is included to demonstrate the tradeoff between mechanical excellence and functional feasibility.

Material	E (GPa)	α ($\times 10^{-6}/\text{K}$)	σ_y (MPa)	K_{IC} (MPa $\sqrt{m}$)	ρ (kg/m ³)	Cost (\$/kg)
Borosilicate Glass	64	3.3	40	0.8	2230	2.0
Fused Silica	72	0.5	20	0.7	2200	2.5
Soda-lime Glass	70	9.0	35	0.75	2500	1.0
Polycarbonate (PC)	2.5	70	65	3.0	1200	2.5
Polysulfone (PSU)	2.5	55	55	2.0	1250	15.0
PEEK	4.0	50	90	4.0	1320	45.0
Alumina (Al ₂ O ₃)	380	8.0	300	3.0	3950	20.0

Table 1: Property values used for candidate material screening.

Notes: Property values were primarily obtained from MatWeb (<https://www.matweb.com>). For properties not available on MatWeb, typical values from publicly available sources were used. Yield strength for brittle materials refers to flexural strength. Thermal expansion coefficients are average values over the range 20–100°C. Costs are estimated based on bulk engineering-grade supply prices.

5. Ashby Chart Interpretation and Selection

The performance indices derived previously are now evaluated graphically using Ashby material selection charts. Each chart plots two key properties and includes overlaid constraint lines or index contours that help narrow the material choices. The results are interpreted in light of the constraints established in Section 2.

Thermal Stability ($M_1 = E/\alpha$)

Figure 1 shows Young’s modulus (E) versus thermal expansion coefficient (α). A red horizontal line indicates the maximum acceptable expansion coefficient of $10 \times 10^{-6}/\text{K}$. Only materials below this line satisfy the thermal stability constraint. Fused silica and borosilicate glass lie well within the acceptable region, with fused silica being the most thermally stable candidate. In contrast, polymers such as PC, PSU, and PEEK lie far above the line due to their high thermal expansion and low stiffness, making them less suitable where dimensional stability is critical.

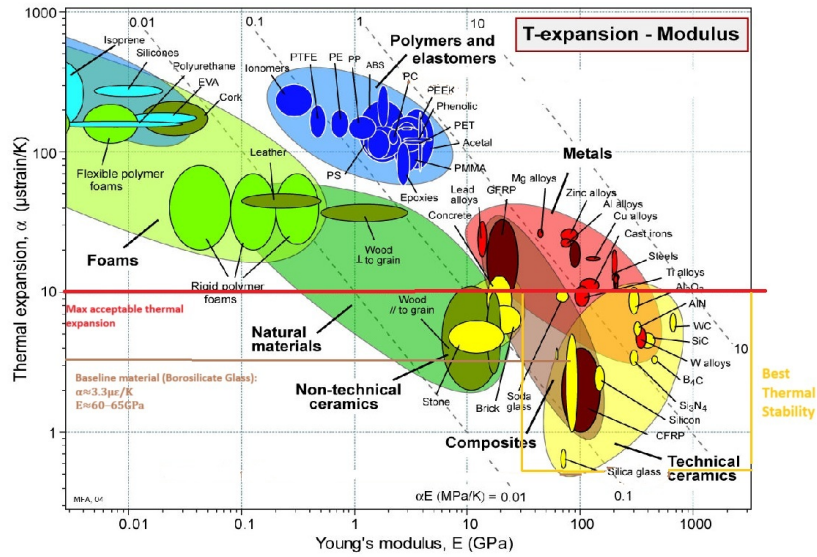


Figure 1: Ashby chart for E vs. α (Thermal Stability Index $M_1 = E/\alpha$). The red line marks the upper limit of acceptable thermal expansion.

Strength-to-Weight ($M_2 = \sigma_y / \rho$)

Figure 2 shows strength versus density, with contours of constant strength-to-weight ratio. Lightweight polymers like PC and PEEK show superior performance in this chart, while borosilicate and soda-lime glasses are heavier and offer moderate strength. Alumina, although very strong, is extremely dense and thus scores poorly by this index.

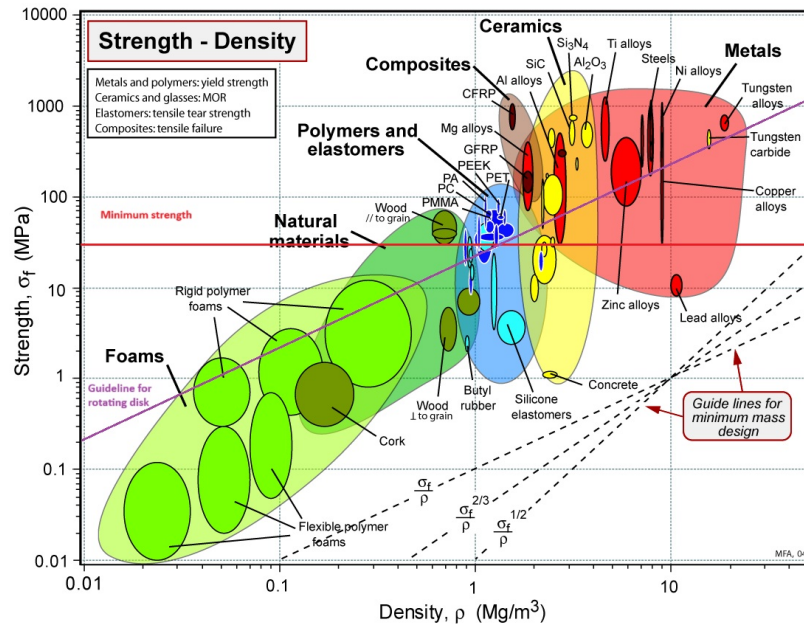


Figure 2: Ashby chart for σ_y vs. ρ (Strength-to-Weight Index $M_2 = \sigma_y/\rho$).

Impact Resistance ($M_3 = K_{IC}^2/E$)

Figure 3 plots fracture toughness (K_{IC}) against stiffness (E). Diagonal dashed contours represent constant values of the impact index. Polycarbonate (PC) and PEEK are located in the upper left, indicating excellent fracture resistance due to their high toughness and low modulus. In contrast, fused silica and soda-lime glass fall below the threshold of acceptable impact performance, consistent with their known brittleness. Borosilicate glass lies on the borderline, offering adequate resistance but not ideal performance in drop scenarios.

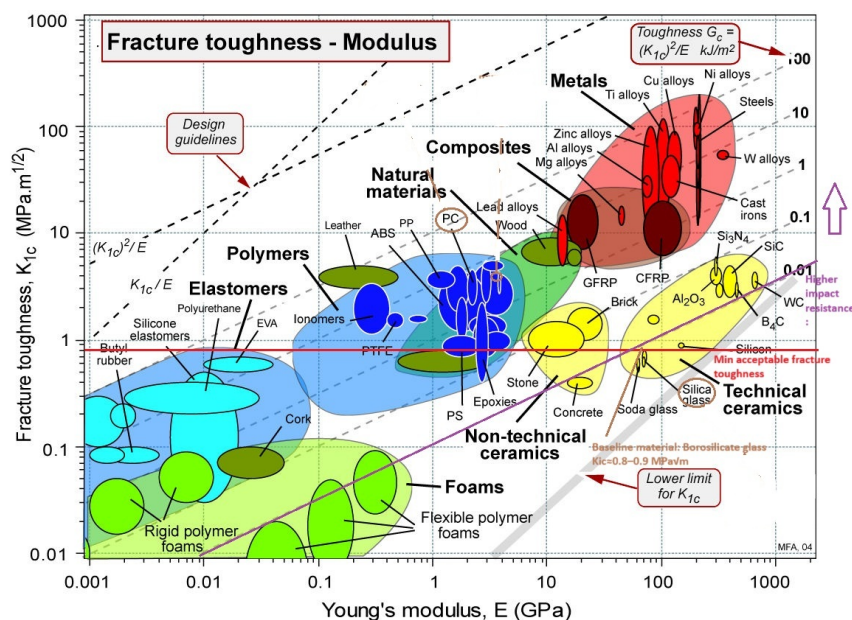


Figure 3: Ashby chart for K_{IC} vs. E (Impact Resistance Index $M_3 = K_{IC}^2/E$).

Thermal Shock Resistance ($M_4 = K_{IC} / E\alpha$)

This index is not represented on a single chart, as it combines three variables. However, it can be evaluated by cross-referencing the E- α and K_{IC}-E plots. Borosilicate glass offers a favorable balance: low expansion, moderate stiffness, and just enough toughness to survive sudden thermal gradients. Fused silica performs best in thermal stability but lacks toughness. PC and PEEK, though tough, expand too much under thermal loads and may warp or crack when subjected to rapid temperature changes. **Cost-Performance** ($M_5 = \sigma_u / C$)

Figure 4 compares strength to relative cost per unit volume. Red lines are drawn at $\sigma_y = 40$ MPa (minimum strength) and $C = 3$ \$/kg (maximum cost). Polycarbonate and borosilicate glass lie in the compliant region, offering sufficient strength within cost limits. PEEK and PSU are excluded due to cost, even though they offer better strength. Alumina also violates the cost constraint and is dense and brittle, which makes it unsuitable for this application.

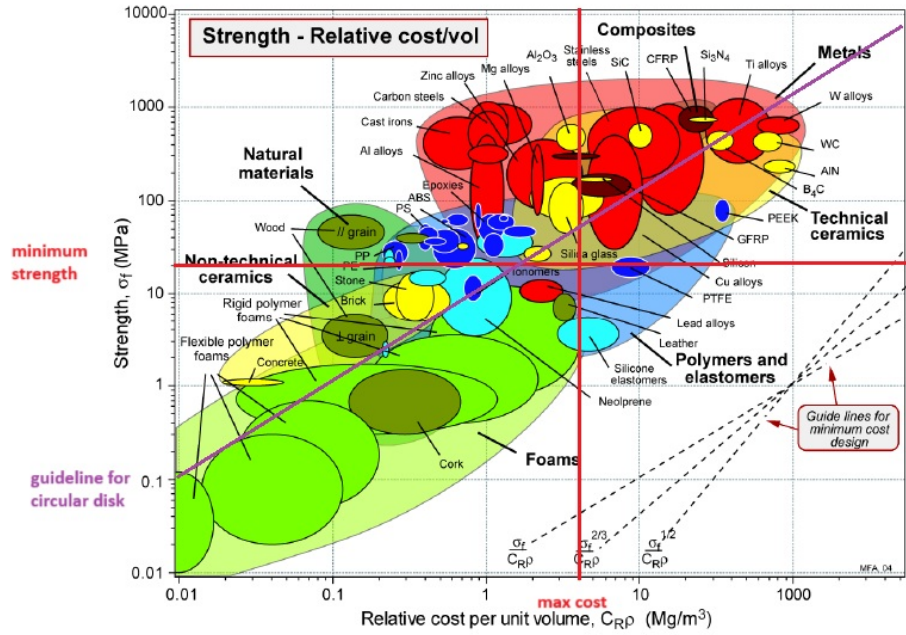


Figure 4: Ashby chart for σ_y vs. cost (Cost-Performance Index $M_5 = \sigma_y/C$).

Additional Candidates Identified from Charts

While the initial screening was based on a preselected set of common materials, a closer examination of the Ashby charts reveals a few additional candidates that may be worth considering. Notably, polymers such as PMMA (acrylic) and epoxy composites appear in acceptable regions across multiple plots. PMMA offers moderate stiffness and strength at very low cost and appears in the compliant region of the cost-performance and strength-to-weight charts. However, it has relatively high thermal expansion and moderate toughness, which could limit its use under thermal shock. Epoxy composites (particularly glass-filled grades) show strong performance in the impact resistance and strength-to-weight plots, but their thermal expansion and microwave transparency need to be carefully validated.

GFRP (glass fiber reinforced polymer) also shows promising behavior in several plots, especially in strength vs. cost and strength vs. density. However, as a composite, it may not offer uniform microwave transparency and can complicate recycling or food-contact compliance.

Despite these findings, none of the alternative materials identified outperform borosilicate glass when all constraints are considered simultaneously. While polymers like PMMA may be cheaper and lighter, they expand too much under thermal cycling. Epoxies and composites may offer higher toughness but raise concerns regarding microwave performance and chemical stability. Thus, the original choice of borosilicate glass remains the most balanced and robust solution for this application.

Taken together, these charts highlight borosilicate glass as the only material satisfying all

critical constraints. PC is a promising backup in designs prioritizing impact resistance, while PEEK and PSU are eliminated due to cost. Fused silica offers the best thermal stability but is disqualified due to fragility.

6. Index-Based Summary and Ranking

To consolidate the results of the previous analyses, Table 2 provides a qualitative ranking of each material for the five indices defined in Section 3. The rankings are based on visual chart interpretation and the numerical property data shown in Table 1. Materials that exceed all performance and constraint thresholds are ranked “Excellent,” while those that marginally satisfy limits are classified as “Moderate” or “Acceptable.” Candidates that fail one or more constraints are marked “Fails.”

Material	M_1	M_2	M_3	M_4	M_5
Borosilicate Glass	High	Moderate	Borderline	Acceptable	High
Fused Silica	Excellent	Poor	Fails	Poor	Moderate
Soda-lime Glass	Moderate	Poor	Poor	Low	Excellent
Polycarbonate (PC)	Poor	Good	Excellent	Moderate	High
Polysulfone (PSU)	Poor	Good	Good	Low	Fails
PEEK	Poor	Excellent	Excellent	Moderate	Fails
Alumina	High	Fails	Fails	Fails	Fails

Table 2: Qualitative ranking of candidates by material index.

The table reinforces several trends observed in the individual Ashby chart analyses. Borosilicate glass ranks acceptable across all indices, particularly for thermal stability (M_1), cost-performance (M_5), and shock resistance (M_4). Fused silica offers unmatched thermal stability but fails on toughness and impact resistance. Polycarbonate performs well in toughness-related indices (M_3), cost (M_5), and weight (M_2), making it a strong backup candidate. PEEK and PSU are eliminated primarily due to cost constraints. Alumina performs well in stiffness alone but is disqualified due to high density, brittleness, and poor thermal shock resistance.

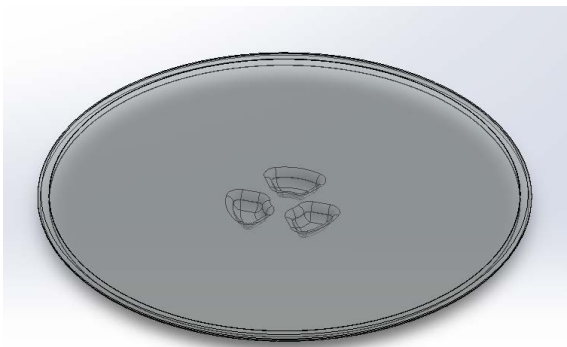
This summary validates borosilicate as the only material that meets all performance and cost requirements simultaneously.

7. Conclusion

This project applied Ashby's material selection methodology to evaluate and compare candidate materials for use in a microwave turntable plate. Beginning with functional requirements—thermal stability, mechanical integrity, food safety, microwave transparency, and cost-effectiveness—we derived five material indices and used Ashby plots to visualize material performance under various constraints. Borosilicate glass, the current industry standard, emerged as the most balanced and reliable choice. It offers a strong combination of low thermal expansion, sufficient mechanical strength, moderate toughness, and low cost. Alternative candidates such as fused silica excel in thermal stability but fail on fracture toughness, while high-performance polymers like PEEK offer superior toughness but are disqualified on cost. Polycarbonate represents a viable secondary option in applications where impact resistance is prioritized over thermal stability.

Additional materials observed from the Ashby charts—such as PMMA, epoxy, and GFRP—show potential in isolated performance categories, but ultimately fall short when all functional and safety requirements are considered together.

In conclusion, while several materials offer tradeoffs in specific areas, borosilicate glass remains the optimal solution when evaluated holistically. Its balanced properties, proven food safety, and compatibility with microwave heating environments justify its continued use in this application.



Isometric View Microwave Plate



Top View Microwave Plate

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